

Muhammad Faizan Khan

Research Interests: Time Series Forecasting | Large Language Models & RAG | Deep Learning | AI Engineering | Computer Vision | Signal Processing | Data Science

Gender Male | **Nationality** Pakistan

School of Computer Science and Technology
National Key Laboratory of Human-Machine
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CAREER OBJECTIVE

PhD researcher at the School of Computer Science and Technology and the National Key Laboratory of Human-Machine Hybrid Augmented Intelligence, Xi'an Jiaotong University, with a Master's degree in Computer Science & Artificial Intelligence from Xidian University. Research covers deep learning for time series forecasting — including energy, biomedical and human activity signals — and Large Language Model based knowledge and retrieval systems. Experienced in developing deep learning, machine learning and signal processing solutions for real-world applications, and in building and deploying AI services with Python, PyTorch, TensorFlow, Scikit-learn and FastAPI. Published as first author and co-author in AI, wireless communications and interdisciplinary journals. Seeking opportunities in AI Engineering, Applied Research and Data Science.

EDUCATION

2025 - Present	Xi'an Jiaotong University, Xi'an, China	PhD, School of Computer Science and Technology	Ongoing
2022 - 2025	Xidian University, Xi'an, Shaanxi, China	Master in Computer Science & Artificial Intelligence	GPA: 88%

DOCTORAL RESEARCH

2025 - Present	School of Computer Science and Technology National Key Laboratory of Human-Machine Hybrid Augmented Intelligence Xi'an Jiaotong University, Xi'an, China
Ongoing Projects	● Multi-horizon wind power forecasting with deep learning models ● Benchmarking and reproduction of state-of-the-art time series Transformers ● Physics-guided forecasting architecture for renewable energy ● LLM Wiki — a maintained knowledge base for large language models ● GraphRAG and multimodal retrieval-augmented generation ● Large language models for time series forecasting ● Contactless respiration monitoring with IR-UWB radar — breathing-signal time series and deep learning ● Human activity recognition from time series sensor data
Industry Prototype	Building an engineering prototype for a client-facing wind power forecasting project, with regular progress reviews. Also present papers at bilingual (Chinese/English) group reading meetings and prepare technical reports and reproduction studies.

PUBLICATIONS

Journal Article	<i>Green HRM and STARA Capabilities in Higher Education: The Mediating Roles of Green Innovation and Green Passion in Sustainable Performance</i> — International Journal of Sustainability in Higher Education (Emerald), 24 August 2026, pp. 1-35. With M. W. Nazir, Z. Shen and L. A. Rahoo. https://doi.org/10.1108/IJSHE-01-2026-0113
Conference Papers — First Author	● <i>Hybrid Modulation Classification: Traditional Signal Processing and LSTM</i> — ICMMT 2025 ● <i>Automatic Modulation Classification using a Deep Learning Model based on ResNet</i> ● <i>A Convolutional Neural Network and Particle Swarm Optimization Hybrid Model for Scripting Language Handwritten Character Recognition</i>
Conference Papers — Co-Author	● <i>Accurate Detection of Heart Disease Using the CardioFusion Ensemble Algorithm</i> — ICECE 2024 ● <i>Generating Synthetic Chest X-ray Images for Improved COVID-19 Detection Using Generative Adversarial Networks</i> — UCOM 2024
Thesis	Master's Thesis: <i>Enhancing Automatic Modulation Classification Using Multi-Stream Deep Learning Architectures</i> , Xidian University, 2025.

PROFESSIONAL EXPERIENCE

Research Assistant & Academic English Writing Specialist	Freelance support in academic writing, research editing and technical proofreading across biomedical sciences, chemistry, AI and technology. Structure scientific manuscripts, improve clarity, and align texts with journal submission standards (IEEE, Elsevier, Emerald). Also collaborate on literature reviews, project reports and grant proposals.
Freelance Web Developer — Fiverr & Upwork, 2017 - 2021	● Delivered more than 50 projects for clients across Europe, North America, the Middle East and Asia over four years on both platforms. ● Designed, built and deployed the institutional website for Government College of Technology (GCT), Faisalabad, including responsive front-end, content management and hosting setup. ● Custom websites and landing pages, WordPress themes, PHP/MySQL backends, e-commerce and CMS setup, site migration, performance optimisation and ongoing maintenance. ● Managed the full client cycle independently: requirements, quoting, delivery to deadline, revisions and post-launch support across time zones.
AI & Backend Development	● Build REST APIs and model-serving endpoints with Python and FastAPI, with Pydantic validation, asynchronous endpoints, background tasks and OpenAPI documentation. ● Deploy machine learning and LLM inference services containerised with Docker and served with Uvicorn; integrate vector databases and retrieval pipelines for RAG applications.
Techlogix Pakistan	Eight-week internship with the Banking Implementation Team on the Flexcube project at Askari Bank. Backend development in Node.js and C#, Flexcube integration in Java, and testing of banking modules. Exposure to MVVM architecture, WPF/WCF applications, real-time forex systems and SQL database management.
NADRA Pakistan	Six-week internship with the Technical Support Team, working with NADRA's proprietary NTS system, MongoDB database management and application development in Node.js.

TECHNICAL SKILLS

Programming	Python (primary), MATLAB, Java, C#, Node.js / JavaScript, PHP, SQL
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AI & Machine Learning	PyTorch, TensorFlow, Keras, Scikit-learn, LightGBM, XGBoost, OpenCV, Hugging Face Transformers, LangChain, FAISS
Research Areas	Time series forecasting, large language models, RAG and GraphRAG, multimodal retrieval, computer vision, natural language processing, signal processing, automatic modulation classification, federated learning, biomedical signal processing, respiration sensing, human activity recognition
Web & Systems	FastAPI, REST API design, Uvicorn, Docker, WordPress, Git and GitHub, Linux, GPU compute platforms
Tools	LaTeX and Overleaf, draw.io, Multisim, Proteus, Microsoft Word, Excel and PowerPoint

CERTIFICATIONS

● Machine Learning — Stanford University (Coursera) ● Neural Networks and Deep Learning — DeepLearning.AI (Coursera) ● CS50's Introduction to Artificial Intelligence with Python — Harvard University
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PERSONAL SKILLS & ACTIVITIES

Communication	Proficient in articulating complex research concepts, methodologies and results. Experienced in authoring peer-reviewed papers, preparing technical reports, and presenting bilingual (English/Chinese) technical talks at group and conference level.
Leadership & Volunteering	● Punjab Youth Festival Ambassador, Faisalabad Region ● IEEE University Chapter, Volunteer ● Vice Manager, University Tech Fair ● Volunteer, International Student Organization, Xidian University

ACHIEVEMENTS & LANGUAGES

Achievements	Awarded the fully-funded Chinese Government Scholarship (CSC) for Master's studies.
Languages	English, Urdu, Pashto and Hindi — fluent. Chinese — intermediate, HSK-3 (264/300).

REFEREES

Prof. Shuyuan Yang — International Research Center of Intelligent Perception and Computation, School of Artificial Intelligence, Xidian University, Xi'an, China. syyang@xidian.edu.cn
Prof. Zhixi Feng — International Research Center of Intelligent Perception and Computation, School of Artificial Intelligence, Xidian University, Xi'an, China. zxfeng@xidian.edu.cn
Dr. Saeed Ur Rahman — Associate Professor, School of Electronic Engineering, Xidian University, Xi'an, China. saeed@xidian.edu.cn